

Assignments-1

(1) Name 5 IT job roles and write one line about what each role does.

Ans:- **Software Developer** – Designs, writes, tests, and maintains computer programs and applications.

Network Administrator – Manages and maintains an organization's computer networks and internet connections.

Database Administrator – Organizes, secures, and manages databases to ensure data is available and accurate.

Cybersecurity Analyst – Protects computer systems and networks from cyber threats, attacks, and unauthorized access.

IT Support Specialist – Helps users troubleshoot and resolve hardware, software, and technical issues.

(2) List 3 product companies and 3 service companies with one example of work each does.

Ans:- **Product Companies**

Microsoft – Develops software products like Windows and Microsoft Office.

Adobe – Creates creative software such as Photoshop and Premiere Pro.

Google – Builds products like Google Search, Gmail, and Android.

Service Companies

Tata Consultancy Services – Provides software development and IT consulting services for clients.

Infosys – Offers application development, maintenance, and digital transformation services.

Accenture – Delivers technology consulting and business process outsourcing services.

(3) Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans:- **GitHub**: GitHub is a platform that helps developers store and manage their code online. It also allows multiple people to work on the same project and keep track of changes.

Slack: Slack stands for *Searchable Log of All Communication and Knowledge*. It is a communication tool used by teams to send messages, share files, and collaborate on projects.

Jira: Jira is a project management and issue-tracking tool commonly used in software development. It helps teams organize tasks, track bugs, and monitor project progress.

IDE (Integrated Development Environment): An IDE is a software application that provides tools for writing, editing, testing, and debugging programs in one place, making software development easier.

API (Application Programming Interface): An API is a set of rules that allows different software applications to communicate and exchange data with each other.

(4) Name any 5 programming languages and mention one popular use case for each.

Ans:- **Python:** Python is a simple and easy-to-learn programming language. It is widely used for data analysis, artificial intelligence, and machine learning applications.

Java: Java is a popular programming language known for its portability. It is commonly used for developing Android mobile applications.

C++: C++ is a powerful programming language that provides high performance. It is often used for game development and system software.

JavaScript: JavaScript is mainly used to make websites interactive and dynamic. It helps create features such as animations, forms, and real-time updates on web pages.

C#: C# is a modern programming language developed by Microsoft. It is widely used for creating desktop applications and video games using the Unity game engine.

(5) What is the difference between a developer and a tester? Answer in 2 lines.

Ans:- **Developer:** A developer writes and maintains software code to create applications and features.

Tester: A tester checks the software for bugs, errors, and quality issues to ensure it works correctly before release.